

1 Basics

$$x, x', x_1, x_n, x_{2n-1}, \hat{x}, \widehat{x}, \mathbf{x}$$

$$y, y', y_1, y_n, y_{2n-1}, \hat{y}, \widehat{y}$$

2 WildCards

$$q, q_a, q_{a,b}, \text{ but accidentally } q_{a+b}$$

$$f, f(a), f(q), f(a, b), \text{ but } f(a + b)$$

$$g, g(a), g(a)(b)$$

$$z(a + b), \widehat{z}(a)$$

3 Practical

$$f(x) = \sum_{i=0}^{\infty} f_i x^i$$

4 Impractical

$$a, b, c, ab, bc, abc$$

$$\varepsilon, \varepsilon_1, \mathcal{T}_\varepsilon, \mathcal{T}_{\varepsilon_1}, \mathcal{T}_{\varepsilon_1, \varepsilon_2, \varepsilon_3}$$

5 Declaration order

$$\varepsilon, \varepsilon_1, \mathcal{T}_\varepsilon, \mathcal{T}_{\varepsilon_1}, \mathcal{T}_{\varepsilon_1, \varepsilon_2, \varepsilon_3}$$

6 Notations?

$$a + b; \mathbf{baz}, \mathbf{foo}, \mathbf{foo}$$

$$\left| (a)_j \right|; \left| (abc)_j \right|$$

$$\left([a]_j \right); \left([abc]_j \right)$$

7 More detail

$$f(x) = \sum_{i=0}^{\infty} f_i x^i$$

8 Euler's Identity: Backporting Semantics

$$e^{i\pi} + 1 = 0$$